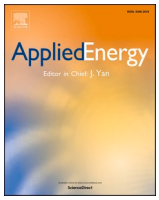
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Reframing Data Centers as Strategic Producers: Energy-Data Nexus

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HIGHLIGHTS

- Energy-Data Nexus reframes data centers from passive loads to strategic producers.
- Data-to-Energy computation reduces grid reserve requirements by lowering uncertainty.
- Verifiable Green Computation Units turn clean energy into a tradable asset.
- Stranded renewables monetized by exporting computation via fiber, not electricity.
- Proof of Green Work underpins a digital carbon border for imported computation.

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ABSTRACT

The exponential rise of artificial intelligence reveals a structural mismatch between two critical infrastructures: digital compute expands on yearly cycles, while power grids evolve on decadal planning horizons. This structural divergence is straining the prevailing passive-consumer model of data centers, resulting in connection queues, carbon lock-in, and de facto moratoria on new facilities. Here, we propose the Energy-Data Nexus, a framework that reframes data centers from passive consumers to strategic producers. The framework defines a three-phase evolutionary pathway built on three operational mechanisms: (1) Energy-to-Data, where data centers act as a "digital battery" by aligning flexible workloads with renewable availability; (2) Data-to-Energy, where computational optimization reduces the requirement for physical grid reserves by lowering operational uncertainty; and (3) Energy-Data Symbiosis, which "liquefies" stranded renewable energy into the proposed Verifiable Green Computation Units, tradable digital assets exported via fiber optics. This paradigm shift not only relieves grid stress but also enables resource-rich regions to monetize stranded renewables by exporting computation rather than electricity, opening a new export pathway within the digital economy.

1. Introduction

Rapid digitalization has precipitated an exponential surge in computational demand. Global data center electricity consumption is projected to double, from 415 terawatt-hours (TWh) in 2024 to nearly 945 TWh by 2030, reaching 3–4% of global electricity demand [1]. This trajectory exposes a fundamental incompatibility between two critical infrastructures: digital compute expands on yearly innovation cycles, while the power grid operates under regulated, capital-intensive planning horizons measured in decades. This temporal divergence means that the power grid is unable to keep pace with digital demand, manifesting in acute spatiotemporal mismatches. The grid's inability to accommodate hyperscalers has resulted in systemic failures: multi-year connection queues, reliance on carbon-intensive stopgap generation, and de facto moratoria on new facility construction.

Digitalization is increasingly recognized as a driver of the energy transition itself: digital tools and sector coupling enable the coordinated, integrated energy systems on which low-carbon urban energy depends [2]. Responses to date, from grid capacity expansion to spatial workload relocation, remain constrained by a "passive consumer" orthodoxy that treats data centers merely as inelastic loads or, at best, marginal demand response participants. As digital demand continues to outpace grid expansion, this model is approaching its operational limits. A growing body of literature has repositioned data centers as flexible grid

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participants. Google's Carbon-Intelligent Computing System has demonstrated production-scale temporal workload shifting to follow low-carbon grid hours [3], while Meta's Carbon Explorer framework jointly optimizes renewable procurement, battery investment, and workload flexibility for 24/7 carbon-free operation [4]. In parallel, geographic load redistribution has been used to spatially route computation toward renewable-rich regions [5]. These efforts harness workload flexibility but still treat data centers as passive loads: they respond to grid signals as electrical consumers rather than reshaping grid operation as strategic producers. The field lacks a unified framework that elevates computation to an active grid infrastructure and a tradable carrier of clean energy. To address this gap, this paper proposes the Energy-Data Nexus (EDN) framework, in which computation reduces the grid's reserve requirements by lowering operational uncertainty, and green computation is verified, traded, and governed as a distinct asset class, as shown in Fig. 1.

The remainder of this paper is organized as follows. Section 2 diagnoses the structural conflicts of the passive-consumer model (Phase I): chronological divergence, capacity hard stops, and spatial disconnects. Section 3 introduces the Energy-Data Nexus framework and its three operational mechanisms: Energy-to-Data (E2D), Data-to-Energy (D2E), and Energy-Data Symbiosis (EDS). Section 4 examines the prosumer transition (Phase II) and the institutional and cyber-physical barriers it confronts. Section 5 develops the strategic-producer phase (Phase III), introducing Verifiable Green Computation Units, the Proof of Green Work protocol, and the governance questions raised by a digital carbon border. Section 6 concludes.

2. Phase I: system conflicts and passive load

In this foundational phase, data centers operate as the passive consumer, an inelastic load that extracts energy from the grid without reciprocal interaction. This model is approaching a physical ceiling: the spatiotemporal mismatch between digital dynamics and grid physics strains local grids and compromises electricity affordability for non-data-center users. Three symptoms are diagnosed as follows:

1) **Chronological divergence**: An irreconcilable velocity gap exists between digital demand and local grid capacity. In Virginia, in-state generation, transmission capacity, and imported energy must expand by 150%, 40%, and 150%, respectively, to meet the predicted digital load, an expansion estimated to raise residential electricity costs by $14–$37 per month by 2040 [6]. The gap forces a regression to fossil fuels. In Tennessee, xAI Memphis Colossus Phase 2 requires 460 MW, which the local grid cannot accommodate [7]. Consequently, the company turned to stopgap modular gas turbines, resulting in a carbon lock-in that directly contradicts sustainability goals.
2) **Capacity hard stop**: In facilities-dense regions, data centers form load clusters that have exhausted grid redundancies. European markets (Frankfurt, London, Amsterdam, Paris, Dublin) have transitioned from stressed to saturated, with grid connection queues averaging 7–10 years [1]. In Ireland, grid constraints produced a de facto pause on new Dublin-region connections, a Google data center was refused planning permission in 2024 on grid-capacity grounds [8], and new connections now require dedicated renewable and dispatchable generation [9].

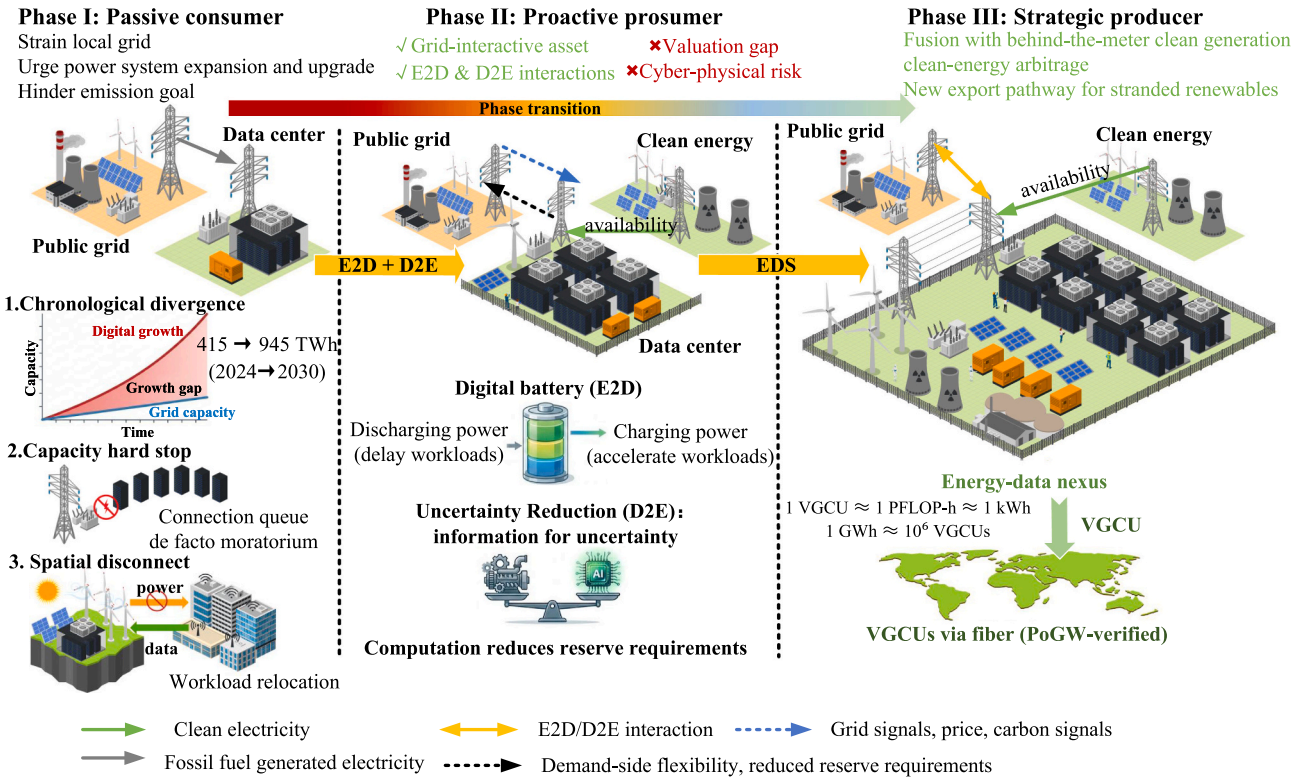


Fig. 1. The structural evolution of data centers within the Energy-Data Nexus. Data centers progress through three phases, driven by the framework's mechanisms. Phase I (passive consumer): one-way consumption strains local grids, producing chronological divergence between digital demand and grid capacity, connection bottlenecks, and spatial disconnects. The E2D and D2E mechanisms drive the transition to Phase II (proactive prosumer): responding to grid signals, renewable availability, and price and carbon signals, the data center provides demand-side flexibility as a digital battery (E2D) and supplies optimization computation that reduces the grid's reserve requirements by lowering operational uncertainty (D2E). The EDS mechanism, fusing computation with behind-the-meter clean generation, drives the transition to Phase III (strategic producer): stranded renewable energy is converted into Verifiable Green Computation Units (1 VGCU ≈ 1 PetaFLOP-hour ≈ 1 kWh) and exported via fiber under Proof of Green Work verification, opening a clean-energy export pathway bounded to the digital economy, while retaining the prosumer-phase grid interactions. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

3) **Spatial disconnect**: Initiatives such as China's "Eastern Data, Western Computing" attempt to resolve the energy conflict via spatial relocation, transferring workloads to renewable-rich regions (e.g., plateau hydro, desert PV with storage) [10]. This improves supply proximity but leaves a fundamental operational disconnect unaddressed in two respects. First, relocation changes the location of the load but not its mode of interaction: the relocated facility remains a passive load, so as western data-center load grows it approaches the same capacity ceiling that constrained the east. The bottleneck is moved geographically, not resolved. Second, relocation aligns computation with where renewables exist but not with when they are available; an uncoordinated load still draws power on its own schedule, forcing fossil backup during renewable shortage and recreating carbon lock-in in a new region. The operational disconnect is therefore the absence of continuous spatiotemporal coordination between workload and grid state, which a one-time spatial move cannot supply.

The fundamental reason for the persistence of the passive-load model is incentive misalignment: active participation imposes private costs on the operator (risk to service-level commitments, operational complexity, and exposure of critical workloads to grid dispatch), while its principal benefits, grid stability and decarbonization, are largely public goods the operator does not capture. For a commercial operator optimizing for uptime and client service, passive operation is the rational default. This misalignment is compounded by three unpriced externalities. Absent a price signal for high-frequency flexibility, the value a data center could provide to the grid goes uncompensated (the valuation gap of Section 4.1). Absent governance of large-load volatility, the stability risk a hyperscale facility imposes goes unaccounted for (Section 4.2). Absent a carbon price reaching digital goods, the cheapest emissions response is fuel-switching rather than grid coordination. The passive state persists due to these missing markets and prices. The EDN framework constructs them through computational-flexibility markets, stability governance, and Verifiable Green Computation Units (VGCUs) with a digital carbon border, so that active participation becomes privately rational. Sections 3 to 5 develop these mechanisms.

3. The energy-data nexus framework

To confront the structural incompatibilities diagnosed in Phase I, the EDN framework enables a fundamental functional shift for data centers. By leveraging the bidirectional convertibility between energy (electrons) and data (bits), the EDN guides the evolution of data centers toward the role of strategic producer through three operational mechanisms: Energy-to-Data (E2D), Data-to-Energy (D2E), and Energy-Data Symbiosis (EDS). The framework builds on and extends several established paradigms of data-center–grid interaction, as shown in Table 1. Demand response [11] and carbon-aware workload scheduling [3] already treat data centers as flexible loads that shift consumption in response to grid or carbon signals, and the E2D mechanism extends this established approach. Virtual power plants aggregate distributed resources into dispatchable generation [12], and transactive energy coordinates such resources through market signals [13]; both, however, transact in electricity (MW or MWh). EDN differs in describing a transition from consumer to producer. D2E reframes computation as a means of reducing the grid's reserve requirements by lowering operational uncertainty, rather than merely shifting consumption; EDS treats verified green computation as a tradable, provenance-bearing asset, the VGCU, exported via fiber rather than as electricity transported through transmission networks. EDN's contribution is therefore not the E2D flexibility, which extends prior work, but the producer-side mechanisms: D2E and the VGCU asset class governed by PoGW, which repositions computation as an active carrier of clean energy.

Table 1
Positioning of EDN relative to existing data-center and grid paradigms.

Paradigm	Data-center role	Traded quantity	What EDN adds
Demand response	Flexible load responding to grid signals	Load reduction (MW/MWh)	D2E reserve-requirement reduction; producer transition
Carbon-aware scheduling	Consumer minimizing own carbon footprint	None (self-optimization)	Producer framing; New traded unit (VGCU)
Virtual power plant	Aggregated dispatchable resource	Electricity (MW/MWh)	Output is computation via fiber, not electricity via wires
Transactive energy	Market-coordinated distributed resource	Electricity + price signals	New traded unit (VGCU) + provenance mechanism (PoGW)
EDN (this work)	Consumer → prosumer → producer	VGCU	–

3.1. Workload Flexibility: Which Computation Can Migrate

The EDN mechanisms are designed for computational workloads that can be shifted in time or space, but this flexibility is neither uniform across workload types nor unbounded in scale, as shown in Table 2. Large-scale AI training and batch or offline workloads are delay-tolerant and interruptible through checkpoint-restart, whose overheads modern checkpointing systems reduce to timescales compatible with minute-level grid dispatch. Such workloads are flexible both temporally, allowing deferral to periods of high renewable availability, and spatially, allowing relocation to renewable-rich regions, since no end user awaits their completion in real time; large distributed training is bounded by inter-site bandwidth and migrates between well-connected sites rather than arbitrarily. These workloads constitute the primary resource for the E2D and EDS mechanisms. By contrast, interactive inference must be served upon request arrival and close to users; it offers little temporal flexibility and only limited spatial flexibility through request routing. The EDN's flexibility mechanisms thus apply principally to delay-tolerant workloads.

The scale of delay-tolerant workloads bounds the performance of the EDN framework, and the ratio of delay-tolerant to delay-sensitive load is not publicly reported, remaining an empirical gap. Current data-center load is modeled as essentially flat across all hours [14], reflecting the passive-consumer operation this work seeks to transform, whereas an industry demonstration in Phoenix, Arizona achieved a 25% power reduction for three hours during peak load while maintaining quality-of-service guarantees [15]. The gap between modeled flat operation and demonstrated flexibility indicates untapped potential rather than a technical barrier.

Table 2
Workload flexibility comparison in EDN.

Workload type	Latency tolerance	Temporal flexibility	Spatial flexibility	Role in EDN
AI training	High	High (interruptible via checkpoint restart)	High (inter-site, within bandwidth limits)	Primary: digital battery (E2D), spatial arbitrage (EDS)
Batch / offline inference	Moderate	Moderate (deferrable)	Moderate (relocatable)	Secondary flexible load
Real-time inference	Low	Low (served on arrival)	Low (request routing only)	Largely non-migratable

3.2. E2D: Temporal Alignment

The E2D mechanism aligns data center operations with grid physics. Non-urgent computational workloads, such as AI model training or batch tasks, are subordinated to grid scheduling, responding to renewable intermittency and grid price or carbon signals. Functionally, this transforms the data center into a "digital battery": it absorbs surplus generation and recedes during deficit. "Charging" corresponds to accelerating flexible workloads when renewable power is abundant, increasing consumption to absorb surplus or otherwise-curtailed generation; "discharging" corresponds to temporally deferring or throttling those workloads, or spatially transferring them to idle sites, when power is scarce, reducing consumption to alleviate local stress. The stored quantity is the computational work performed during periods of oversupply. Unlike electrochemical batteries, whose energy carrier degrades chemically with repeated cycling, the digital battery stores energy in the form of computed artifacts and avoids this wear. An important asymmetry nonetheless distinguishes it from electrochemical storage: whereas a physical battery discharges by injecting power back to the grid, the digital battery does not supply electricity; it "discharges" only by reducing its own demand. The analogy thus concerns the temporal and spatial shifting of energy use, not bidirectional power injection. Nor is it cost-free: workload migration introduces performance penalties from checkpointing and restart, communication and bandwidth overhead from relocating computation, and service-quality constraints where deferral or relocation risks violating latency or availability requirements. These costs bound the flexibility the digital battery can offer and distinguish it from an idealized lossless store; they are discussed further as the "penalty paradox" in Section 4.1.

3.3. D2E: Uncertainty Reduction

While E2D offers temporal flexibility, D2E addresses uncertainty. To maintain reliability, grid operators hold operating reserves sized against forecast error, demand variability, and renewable intermittency, traditionally backed by carbon-intensive standby generation such as idling gas turbines. Through the D2E mechanism, data centers apply their parallel computational capacity to high-dimensional forecasting and scheduling under uncertainty, improving operational decisions. Because reserves are sized to cover uncertainty, reducing that uncertainty reduces the quantity of reserves that must be scheduled [16–18].

Computation cannot replace reserve capacity: when generation is lost, only physical assets can inject the required power. D2E's contribution is therefore confined to the timescales on which reserve requirements are set by information rather than physics. Primary frequency response, acting within sub-seconds and governed by system inertia and governor action, lies outside this scope and remains physical. Deterministic contingency reserve sized to the N-1 criterion is likewise unaffected: it is fixed by the largest single contingency, not by forecast quality. D2E's contribution is confined to reserves sized against probabilistic uncertainty on the secondary and tertiary timescales (minutes to tens of minutes), where reserve scheduling is driven by forecast error and improved computation reduces the margin that must be held. Here the data center does not act as reserve itself; it provides the decision-support computation that lets the operator schedule less standby capacity. The energy it consumes is thus not a grid burden but an investment in operational efficiency, reducing reliance on continuously emitting standby generation without compromising the physical reserves reliability requires.

The computational demand of D2E arises specifically from scheduling under uncertainty. Relevant methods include scenario-based stochastic and robust unit commitment and economic dispatch, optimal power flow, model predictive control for receding-horizon redispatch, and learned surrogates for real-time operation; the appropriate choice is system- and objective-dependent. Such computation is modest relative to the data center's primary workload. As an order-of-magnitude reference, solving a demanding dense linear system (34,560 × 34,560, representative of the linear-algebra core of power-flow and dispatch computation) consumes an estimated 5–30 MJ (approximately 1.4–8.3 kWh) over minutes on high-performance hardware [19], whereas a single large-model training run such as GPT-3 requires on the order of 1287 MWh [20], five to six orders of magnitude greater. Even the added cost of representing uncertainty leaves grid optimization far below the scale of AI training, so a data center can supply it as a marginal activity rather than through dedicated resources, at an overhead small relative to the continuous emissions of the standby generation it helps displace.

Interpretability is preserved by the nature of the computation: D2E solves classical, constraint-based optimization that is mathematically transparent and auditable, not opaque neural-network inference, and the AI-capable hardware is incidental to the method. The data center functions as decision support, supplying dispatch setpoints, reserve schedules, and commitment decisions [21] that the grid operator verifies against security constraints and may accept or reject, retaining full authority; learned surrogates accelerate the solving process and their outputs are checked against feasibility constraints before use, serving as auditable accelerators rather than autonomous controllers. Integration draws on standard inputs such as SCADA/EMS telemetry, renewable and weather forecasts, and network and market data [22], and on established utility communication standards including IEC 61850 and ICCP/TASE.2 [23,24]. The detailed protocol design, and a full accounting of optimization overhead against avoided reserve emissions, remain implementation-stage questions that the EDN framework is intended to motivate.

3.4. EDS: Cyber-Physical Symbiosis

The EDS mechanism drives the data center toward the final phase: the strategic producer. The data center is integrated with behind-the-meter clean generation (e.g., small modular reactors or gigawatt-scale solar), creating the autonomous physical entity the EDN framework envisions. This cyber-physical symbiosis enables the production of VGCUs. A VGCU is a tradable digital asset representing a standardized, benchmarked quantity of useful computation, cryptographically certified via the Proof of Green Work (PoGW) protocol (proposed in Section 5.3) to have been produced using verified renewable electricity at a specific time and location. Because delivered computation depends on numerical precision, utilization, and hardware architecture, a VGCU is denominated not in nominal peak FLOPs but in a standardized unit of effective computation measured against reference benchmark workloads at a specified numerical precision, analogous to how MLPerf-class benchmarks normalize throughput across heterogeneous accelerators (e.g., NVIDIA and Huawei hardware) [25]. One VGCU therefore corresponds to a benchmarked quantity of useful compute, rendering units fungible across different platforms.

Functionally akin to liquefied natural gas (LNG), VGCUs "liquefy" stranded renewable energy into globally tradable computation. The term is analogical, denoting the conversion of geographically stranded energy into a transmissible product, not a physical phase change. This avoids the cost and political hurdles of physical energy transport: value is exported via fiber optics rather than as electricity through high-voltage transmission. VGCUs enable a form of clean-energy arbitrage bounded to the digital economy: resource-rich but geographically isolated regions can monetize otherwise-stranded renewable energy by exporting computation via fiber rather than transmitting electricity. For such regions, the value lies in converting renewables that would otherwise be inaccessible to distant markets into a tradable computational product; for importing regions, in access to verified green computation. This extends energy-derived economic participation into the computational domain, rather than redefining energy sovereignty in the broad sense associated with general-purpose fossil fuels.

Although a VGCU certifies computation produced with clean energy, it is categorically distinct from existing instruments such as Renewable

Energy Certificates (RECs) [26] and Guarantees of Origin (GoOs) [27]. A REC represents the non-power environmental attributes of 1 MWh of renewable generation and can be unbundled and traded separately from the electricity; a GoO is its European equivalent under the EU Renewable Energy Directive. Both certify generation, are denominated in energy (MWh), and are typically matched only on an annual or monthly basis. A VGCU instead certifies that a specific quantity of useful computation was performed using renewable energy, with PoGW binding the computation and the generation to a common time and grid node. The distinction is one of matching scope: RECs and GoOs attest that renewable energy was generated at some time and place, decoupled from any specific use, whereas a VGCU attests that a specific computational output was produced under verified spatiotemporal coincidence of compute and clean generation as shown in Table 3. This bundling of the green attribute to delivered useful work, together with enforced spatiotemporal simultaneity, is what constitutes VGCUs as a new asset class rather than a repackaging of existing instruments.

To illustrate that the VGCU is an operationalizable unit, consider a concrete example. We denominate one VGCU as one PetaFLOP-hour of computation at FP16 reference precision, equivalent to 3.6×10^{18} floating-point operations. Taking a current-generation NVIDIA H100 (SXM) accelerator as reference, which delivers approximately 1 PFLOP/s of dense FP16 Tensor Core throughput [28], one VGCU corresponds to approximately one accelerator-hour of useful computation. Including server overhead (CPU, memory, networking) and facility cooling at a power usage effectiveness (PUE) of 1.2, the accelerator's approximately 700 W draw implies that producing one VGCU consumes on the order of 1 kWh of verified renewable electricity. This yields a transparent relationship linking standardized computation directly to physical energy: 1 VGCU $\approx$ 1 PetaFLOP-hour (dense FP16) $\approx$ 1 kWh of clean electricity. At this order of magnitude, 1 GWh of stranded renewable generation maps to roughly one million VGCUs, lending concrete scale to the energy-arbitrage potential of the EDS mechanism. The PUE assumption is deliberately conservative: leading hyperscale operators report annual PUE below 1.11 [29], while the industry average remains near 1.5–1.6 [30], so 1.2 reflects an efficient modern facility without assuming best-in-class performance. These figures are necessarily illustrative, as actual conversions depend on workload, utilization, numerical precision, and cooling efficiency. Their purpose is to establish that the unit is well-defined and operationalizable, not to fix a precise constant.

4. Phase II: proactive prosumer and bidirectional interaction

The phase transition from a passive consumer to a prosumer is theoretically feasible under the proposed EDN framework, but its implementation confronts acute institutional friction. Although the principle of reducing reserve requirements through computation is sound, the sociotechnical infrastructure of current power grids remains maladapted. Two barriers prevent this transition: market incompatibility and cyber-physical vulnerability, as shown in Fig. 2.

Table 3
Renewable energy certificates / guarantees of origin vs. verifiable green computation units.

Dimension	REC / GoO	VGCU
Object certified	Renewable generation	Useful computation performed with renewable energy
Unit of denomination	Energy (1 MWh)	Computation (benchmarked FLOP-hours)
Attribute bundling	Unbundled; tradable separately from the electricity	Bundled to delivered computation
Temporal matching	Annual or monthly	Hourly (24/7), enforced by PoGW
Spatial matching	Generally none (geographically decoupled)	Resolved to grid node
Provenance claim	Renewable energy generated at some time and place	Specified computation produced with verified renewable energy at a given time and node
Double-counting control	Susceptible via unbundling and multiple claims	Prevented by PoGW binding and certificate retirement

4.1. The Valuation Challenge: Market Design Lag

The primary bottleneck is not technical capability, but the market design lag for digital services. Current wholesale electricity markets (e.g., PJM in the US [31], Nord Pool in the EU [32]) are designed for MW or MWh rather than digital assets.

1) **Granularity Mismatch**: Current ancillary service markets typically procure reserve power on relatively coarse time intervals, such as 15 min or hourly, whereas data center flexibility, via direct power modulation of the load rather than workload migration, operates on millisecond-to-second timescales. The power grid's limited ability to price this high-frequency reliability creates a "missing money" problem: data centers providing millisecond-responsive load flexibility are compensated at the rates of slower fossil-fuel generators.
2) **Penalty Paradox**: Under current demand response (DR) rules in most wholesale markets, participants face penalties if they fail to deliver the contracted level of flexibility. If a data center modulates its load to stabilize the grid but fails to meet a rigid baseline due to an unexpected surge in user traffic (e.g., intensive AI queries), this damages client trust and quality of service. This asymmetric risk profile strongly disincentivizes hyperscalers from exposing their critical workloads to grid dispatch signals.

Therefore, a new asset category for computational reserves with an explicit incentive mechanism is a prerequisite for transitioning data centers from passive loads to prosumers.

4.2. The Security Challenge: Cyber-Physical Threat

Integrating data centers with grid physics introduces a profound cyber-physical risk. Under the Phase I passive consumer model, cyber-attacks on data centers, such as email outages or server halts, merely disrupted digital services. However, under the Phase II prosumer model, a data center could be attacked when providing fast-response grid services, and a compromised hyperscaler with GW-scale load could be used to threaten grid stability. Specifically, if clusters of data centers were toggled in sync with wide-area grid oscillation modes, malicious load modulation could potentially trigger cascading blackouts or damage substation transformers, with consequences comparable in scale to the April 2025 Iberian blackout [33].

The cyber-physical coupling between data centers and the grid is bidirectional, and two further risks arise from the data center as a source rather than a target. First, the D2E mechanism requires the grid operator to share confidential operational data, including network topology, real-time state, and generator and contingency information, with the data center performing the optimization. This externalization expands the grid's attack surface, since such data provides a precise roadmap for a targeted attack if disclosed. However, third-party access to sensitive grid data is not unprecedented: Energy Management Systems vendors, planning consultants, and inter-operator exchanges all receive sensitive grid data under established critical-infrastructure security frameworks and contractual data-governance requirements, and confidential-computing environments additionally allow this data to be processed without full disclosure to the data-center operator, applying in reverse the provenance-attestation machinery of Section 5.3. Second, a hyperscale facility constitutes a GW-scale load whose consumption can shift suddenly and stochastically, through traffic surges or the starting and stopping of large workloads. Unlike the diverse, scattered, and

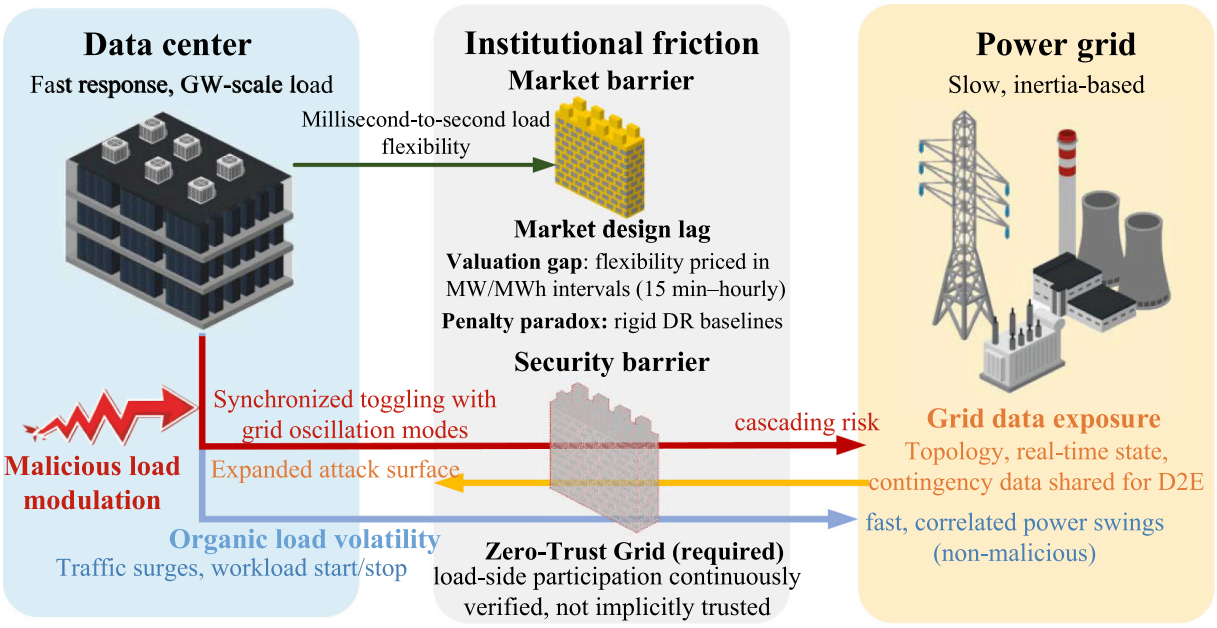


Fig. 2. Institutional friction and bidirectional cyber-physical risk in the Prosumer phase. The transition is hindered by two systemic barriers: (1) market incompatibility, where the grid's inability to price high-frequency load flexibility creates a valuation gap; and (2) cyber-physical vulnerability, which is bidirectional: malicious load modulation threatens the grid through the data center, D2E requires the grid to expose confidential operational data, and organic load volatility from normal hyperscale operation poses a comparable stability challenge. A Zero-Trust Grid architecture (shown translucent), in which load-side participation is continuously verified rather than implicitly trusted, is the barrier this phase requires but current grids lack.

uncorrelated conventional demand, this load is concentrated and may be correlated across facilities responding to a common trigger, producing fast, large power swings that the grid struggles to absorb: sudden load growth depresses system frequency and sudden loss raises it. This load volatility is distinct from the malicious load modulation discussed above: it arises from normal operation, yet poses a comparable challenge to stability. A "Zero-Trust Grid" architecture, in which load-side participation is continuously verified rather than implicitly trusted, must therefore account not only for adversarial manipulation but also for the legitimate operational variability and data-exposure risks that hyperscale participation introduces.

5. Phase III: the strategic producer

In the final phase of EDN evolution, the data center transcends its role as a grid prosumer to become an autonomous strategic producer. By fusing hyperscale compute with behind-the-meter renewables via the EDS mechanism, these facilities unlock a new paradigm of global trade: virtual energy arbitrage through VGCUs. This shift has significant geopolitical and governance implications, opening a new export pathway within the digital economy, as shown in Fig. 3.

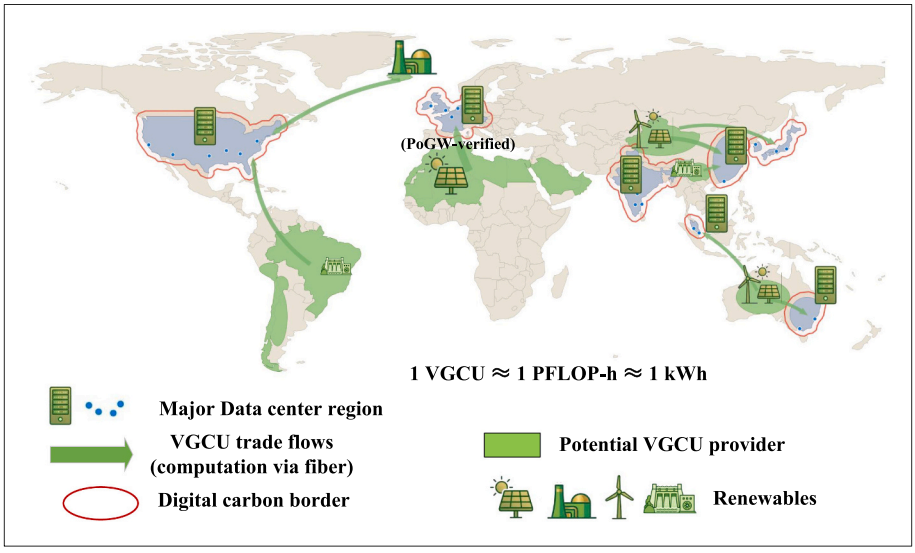


Fig. 3. A new export pathway for stranded renewables. The EDN framework enables resource-rich regions to "liquefy" stranded renewables into VGCUs, exporting value via fiber optics rather than transmission lines. To prevent digital carbon leakage, a digital carbon border enforces the Proof of Green Work (PoGW) protocol, ensuring the environmental integrity of imported computation. Regions shown are indicative, based on [1]. Boundaries are illustrative and imply no position on territorial or jurisdictional questions.

5.1. The Geopolitical Shift: A New Export Pathway for Stranded Renewables

Historically, energy sovereignty has been defined by the physical control of fossil fuels (e.g., oil, natural gas) and the connectivity of high-voltage transmission lines. This model inherently hinders resource-rich but geographically isolated regions from participating in energy markets. For example, Middle Eastern deserts hold abundant solar energy [34] but lie far from major demand centers, where constructing long-distance transmission infrastructure is economically prohibitive.

The EDN framework circumvents these geographic hurdles by "liquefying" stranded renewable energy into VGCUs, exporting high-value computational assets via fiber optics rather than transporting electricity (MW or MWh) through high-voltage transmission networks. This paradigm creates a new export pathway: a state converts domestic renewable resources into globally tradable computational services. It extends economic participation in the digital economy to resource-isolated regions that previously could not access it. This advantage is sectoral, more limited in scope than the general-purpose leverage historically associated with fossil-fuel control. Consequently, the data center acts as a conversion engine, offering resource-rich regions a pathway to convert the historic "resource curse" into a "digital dividend" and reducing the dependence of energy-derived income on physical export infrastructure.

5.2. The Governance Challenge: Stopping "Digital Carbon Leakage"

Current carbon-border policy does not reach digital trade. The EU Carbon Border Adjustment Mechanism (Regulation (EU) 2023/956) [35], for instance, applies under its Annex I only to tangible goods (cement, iron and steel, aluminium, fertilisers, electricity, and hydrogen), leaving computational outputs entirely outside its scope. This creates a regulatory loophole: energy-intensive tasks such as AI model training can be performed in jurisdictions with cheap, high-carbon electricity, and the resulting model parameters imported with zero embodied-carbon accountability. The global fiber network thus risks becoming a leakage channel for embodied carbon.

5.3. Solution to Carbon Leakage: Digital Carbon Border

The realization of Phase III necessitates a digital carbon border. To extend provenance requirements to intangible computational assets, we propose the Proof of Green Work (PoGW) protocol. Whereas Bitcoin's Proof of Work expends energy to demonstrate computational effort, PoGW inverts this logic: it certifies that useful computation was performed using verified clean energy. Concretely, PoGW binds four attributes to each VGCU into a single tamper-evident cryptographic commitment (e.g., a hash) that cannot be altered without detection: a task identifier, a generation timestamp, a physical location (grid node), and the renewable-energy provenance. This commitment is anchored to two independent sources of truth: hardware-attested compute logs generated within trusted execution environments (e.g., Intel SGX, AMD SEV), which verify that the claimed computation physically occurred on certified hardware; and timestamped, location-resolved renewable-generation data drawn from granular certification frameworks such as EnergyTag [36]. That the attested computation and the attested clean generation coincide in time and place is precisely what distinguishes a verified VGCU from an unbundled energy certificate.

VGCU integrity requires that each unit of clean energy be claimed only once. A unit of verified renewable energy carries one environmental attribute, and minting a VGCU is its terminal use: the attribute is retired and can no longer count toward the operator's Scope 2 emissions, preventing a second claim. The provenance requirement depends on configuration. A behind-the-meter facility, integrated with its own generation, establishes provenance directly, since that generation is physically dedicated to its computation. A grid-connected facility, drawing from a shared pool, requires an additional layer: issuance must be backed by renewable certificates matched hourly (24/7) and by location, tied to a specific grid node and interval rather than the annual, geographically decoupled matching of conventional certificates, then retired in a registry upon consumption. PoGW binds each VGCU to its retired, time- and location-stamped certificate; because retirement removes that certificate from the pool, no two parties can claim the same unit. This is the integrity that unbundled, annually matched certificates lack.

This retirement-and-matching logic adapts an established anti-double-counting mechanism: RECs [26], Guarantees of Origin [27], and carbon-credit registries [37] all prevent double-claiming by retiring a serialized unit upon use. PoGW applies this logic at the hourly, location-resolved granularity formalized by the EnergyTag Granular Certificate Standard [36], binding a retired, time- and location-stamped certificate to a unit of computation, and thereby extends a proven accounting mechanism to verified green compute.

The PoGW lifecycle for minting and verifying a VGCU proceeds in six steps. First, a flexible workload is executed on attested hardware drawing verified renewable electricity. Second, the trusted execution environment produces a hardware-attested log recording that the computation occurred on certified hardware and the quantity of energy it consumed. Third, this energy draw is matched against time- and location-resolved renewable-generation certificates for the corresponding interval and grid node. Fourth, the four provenance attributes (task identifier, timestamp, location, and energy source) are bound into a single tamper-evident cryptographic commitment, the PoGW proof. Fifth, the matched generation certificate is retired in a registry, ensuring the same renewable energy cannot be claimed again. Sixth, any party, whether an importer or a regulator at the digital carbon border, can verify the proof against the public commitment, confirming the computation's provenance without access to its contents. This lifecycle renders the verification process concrete and auditable. The full cryptographic specification, including proof-generation overhead and verification latency, lies beyond the scope of this paper and constitutes an important direction for future work; the binding structure described here establishes the minimum informational basis on which trustworthy green-compute trading can rest.

Deploying PoGW at scale faces real obstacles that its design must confront. First, the hardware basis for attestation is emerging rather than universal: current-generation AI accelerators increasingly incorporate confidential-computing capabilities, for example, the NVIDIA H100 provides on-die confidential computing [28], but coverage across older and heterogeneous fleets remains incomplete, so attested execution cannot yet be assumed everywhere. Second, operators have legitimate concerns that disclosing hardware logs could reveal proprietary model details or customer data, which is addressable in principle: attestation can prove a computation's provenance properties (task, time, location, and energy source) without exposing the workload's contents, using privacy-preserving techniques such as zero-knowledge proofs or confidential-computing enclaves that expose only metadata. The design requirement is thus to attest provenance without disclosing payload. Third, proof generation and verification impose computational and communication overhead; for PoGW to be worthwhile, this overhead must remain small relative to the computation certified, a condition favorable for the large workloads VGCUs represent; its precise quantification remains future work. These barriers define the engineering agenda for operationalizing verifiable green computation.

Extending carbon-border logic to computation raises legal questions that PoGW alone cannot resolve. The most fundamental is classification. Computation delivered as processing over a network is most naturally regarded as a service under the General Agreement on Trade in Services, rather than the goods-based frameworks on which carbon border adjustment rests; on this view, a goods instrument such as CBAM cannot reach it by construction [35]. Yet the object that carries embodied carbon across a border, a trained model or computed result transferred

as a persistent, reusable artifact, occupies an unresolved space between service and digital good, a classification contested in international trade law for over two decades. The leakage vector thus falls between goods-based and services-based regulation, and at present is governed by neither. Any provenance-based remedy would further need to satisfy World Trade Organization non-discrimination principles, most-favoured-nation and national treatment [38], under which differentiating comparable computational outputs by their energy source would face a "like products" test and could be contested as a non-tariff barrier to trade in services. These questions lie beyond this paper and require expertise in international trade and climate law. We identify them instead as the governance frontier a digital carbon border must confront. PoGW supplies the technical precondition for such a border, verifiable provenance, but its institutional realization awaits coordinated development in trade and climate governance.

6. Conclusion

This paper diagnoses the structural incompatibility between exponential digital demand and slow grid physics, identifying three systemic failures: chronological divergence, capacity hard stops, and spatial disconnects. To resolve these conflicts, we propose the Energy-Data Nexus framework, which transforms data centers from passive consumers into strategic producers. First, we introduce the Energy-to-Data and Data-to-Energy mechanisms to align operational timescales between energy and data: Energy-to-Data enabling data centers to function as "digital batteries" for flexible workload scheduling, and Data-to-Energy using computation to reduce the grid's reserve requirements by lowering operational uncertainty. Second, two institutional barriers are identified: market design lag and cyber-physical vulnerability, which can be mitigated with new incentives for computational assets and a "Zero-Trust Grid" architecture, respectively. Finally, the framework extends to the Energy-Data Symbiosis mechanism, which converts stranded renewables into Verifiable Green Computation Units. By integrating the Proof of Green Work protocol, this approach establishes a digital carbon border to prevent digital carbon leakage, enabling resource-rich regions to monetize stranded renewables without dependence on physical export infrastructure. The Energy-Data Nexus thus offers a concrete pathway to reposition computation from a resource burden into a strategic asset for global decarbonization and sectoral clean-energy trade. Realizing this vision will require coordinated advances across power systems engineering, market design, cryptographic verification, and international climate governance.

CRediT authorship contribution statement

Xiaoying Shi: Writing – review & editing, Writing – original draft, Visualization, Investigation, Formal analysis, Conceptualization. **Zhenjia Lin:** Writing – review & editing, Investigation. **J. Yan:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Jinyue Yan reports financial support was provided by Hong Kong Research Grants Council. Xiaoying Shi reports financial support was provided by Hong Kong Research Grants Council. Jinyue Yan reports a relationship with Advances in Applied Energy (Elsevier) that includes: board membership. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used Claude (Anthropic) to assist with language editing, manuscript revision, and preparation of reviewer responses. The authors reviewed and edited all content and take full responsibility for the published article.

Data availability

No data was used for the research described in the article.

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